Guideline for
Initial Ventilation Settings for a patient admitted to ICU, or intubated on ICU

Summary

Ventilation can improve gas exchange, reduce work of breathing and improve LV function. However, it is not a cure – it merely 'buys time' while other therapeutic interventions act, or natural disease progression occurs.
As a result, we are not aiming for 'normal' arterial blood gases, we merely want to provide acceptable gas exchange, while minimising any Ventilator Induced Lung Injury (VILI).
Protective ventilation has been shown to be associated with improved outcomes, and these principles should be applied any time you manage a ventilated patient.

It is imperative that you call the duty ICU Consultant immediately in the face of any kind of difficulties with mechanical ventilation.

Key Points

A useful starting point for a **non-wheezy** patient is:

- SIMV (with autoflow) or PRVC
- Tidal volume 6ml/kg ideal body weight (NOT actual body weight). 450ml for adult male, 380 ml for adult female
- Rate 20 bpm.
- I:E 1:1.5.
- PEEP 10cmH2O.
- FiO2 0.8

Connect the patient and check

- Inspiratory pressure the above settings generate (should be <30cmH20)
- Expiratory time allows full emptying (is there any end expiratory flow?). If there is persistence of end expiratory flow, lengthen the I:E to 1:2
- SaO2 92-94% (titrate fiO2 as necessary).

After 10 minutes, check an ABG

- PaCO2 >10kPa, initially increase RR to 24bpm, but make sure you don't gas trap as above. If inspiratory pressures <20cmH20, you can increase Vt to 500ml.
- Titrate fiO2 to PaO2 8-10 kPa.

Asthma and reversible airways disease
In this patient group, gas trapping is the major problem. Allowing sufficient expiratory time for emptying is the key. Mechanically ventilating an asthmatic can be particularly hazardous, and the ICU consultant should always be involved.

- SIMV (no autoflow)
- Tidal volume 6ml/kg
- Rate 12bpm
- I:E 1:3
- PEEP 5cmH2O
- fiO2 0.8

PaCO2

- PaCO2 WILL be high. It will not improve until the bronchospasm abates. Increasing RR in these circumstances will increase gas trapping, increase alveolar dead space and make the patient worse.
- Check expiratory flow pattern. If end expiratory flow persists, lengthen I:E ratio
- Peak airway pressure will increase, as inspiratory time is shortened, you may have to reduce the Vt to keep Pinsp within safe limits.
- In pure asthma, you can reduce PEEP to 2cmH20. In COPD you can't.
- Sometimes you have to disconnect patient from ventilator for ~20s to assist lung emptying and reduce gas trapping.

PUBLICATION DETAILS

Author of Clinical Guideline	Roland Black, Consultant ICM
Directorate/Department responsible for Clinical Guideline	Intensive Care Medicine
Contact details	Ex 2424
Version number	1.1
Replaces version number	1.0
Date written	2015
Approving body and date approved	ICU Governance Group 17-10-18
Review date	17-05-21
Expiry date	17-10-21
Date document becomes live	17-10-18